

CIV102 Project Team 502 Engineering Drawings, Construction Section, and Time Log

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Professor Bentz

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November 23, 2025

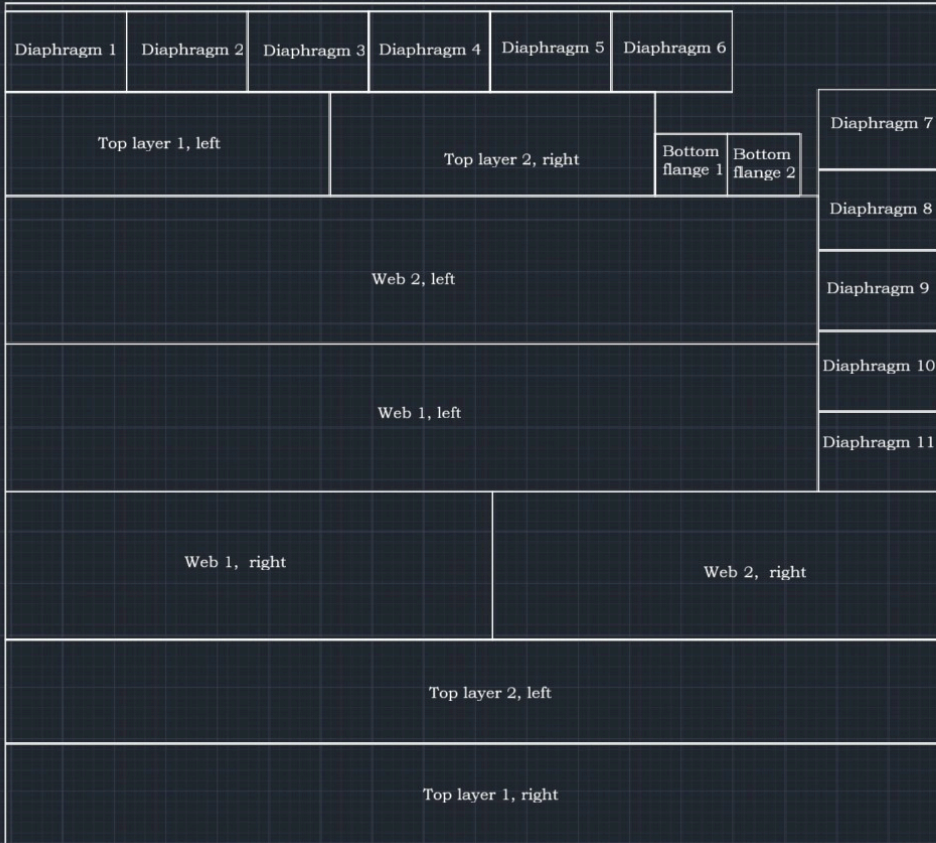
2D cut plan of the matboard

11 diaphragms $77,5 \times 137,5$

2 legs $142,5 \times 1260 \Rightarrow (472,5 + 787,5) \times 142,5$

2 top layer $100 \times 1260 \Rightarrow (315 + 945) \times 100$

2 foot 60×70

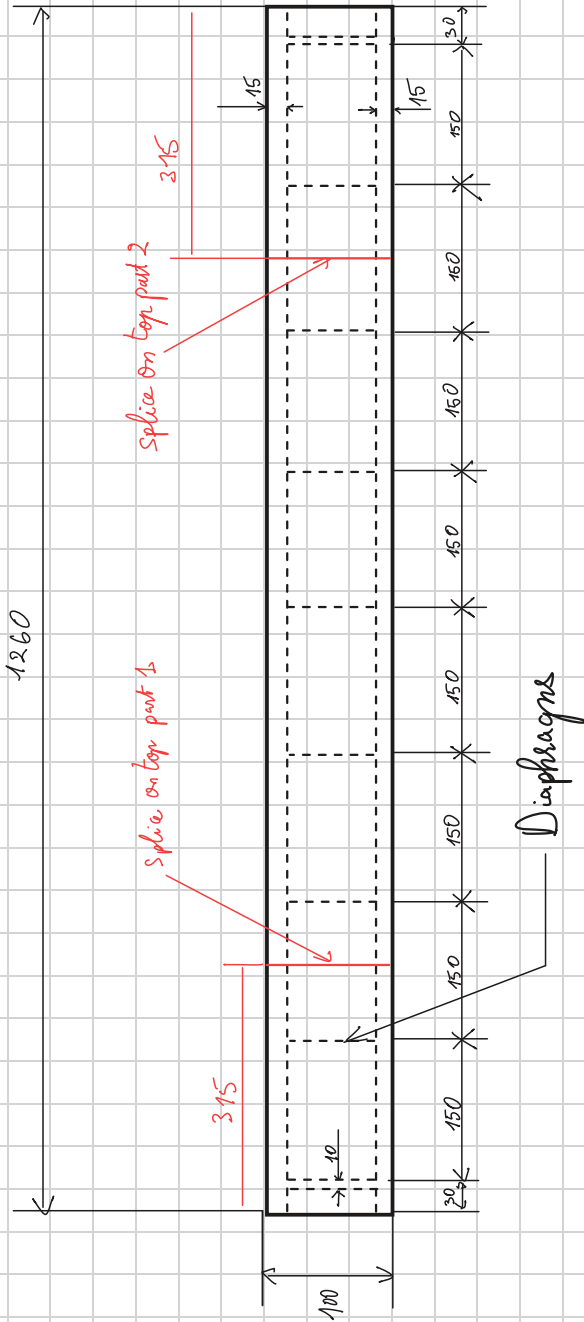


Hand-drawn structural diagram of a beam with dimensions and labels:

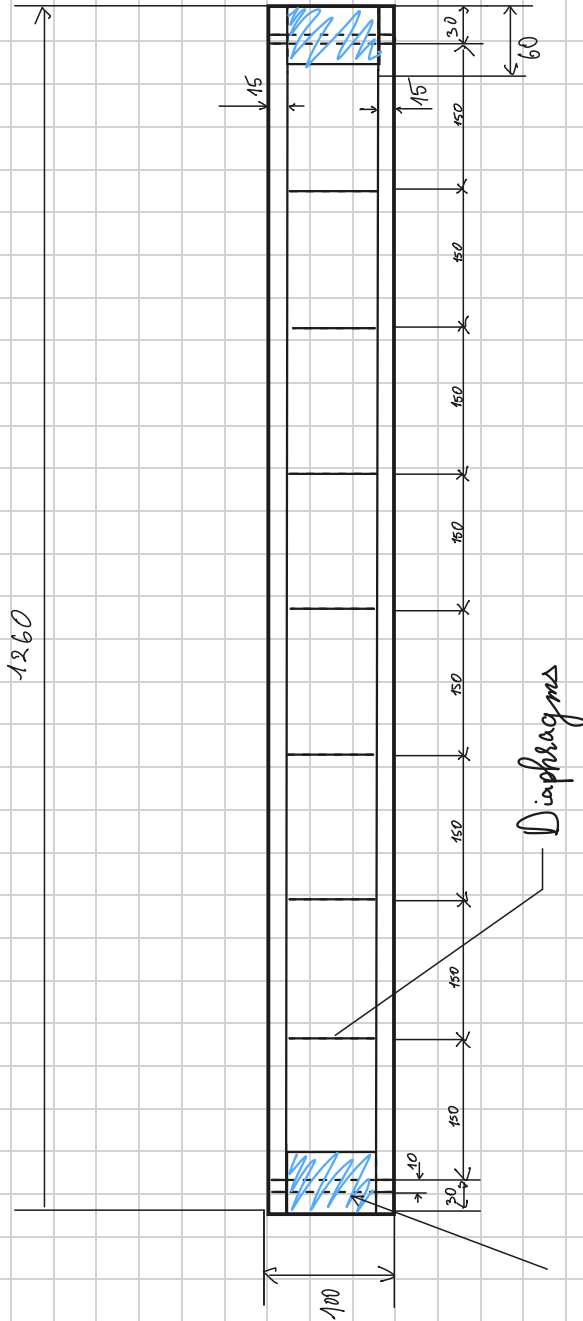
- Overall Length:** 1260
- Overall Height:** 140
- Top Flange Thickness:** 60
- Bottom Flange Thickness:** 30
- Web Thickness:** 10
- Top Flange Width:** 197
- Bottom Flange Width:** 150
- Splice on side 1:** Indicated by a red rectangle and arrows.
- Splice on side 2:** Indicated by a red dashed rectangle and arrows.
- Splice support:** Indicated by a red line and arrows.
- Diaphragms:** Indicated by a line and arrows.
- Bottom web support:** Indicated by a line and arrows.
- Dimensions:**
 - Top flange: 60
 - Bottom flange: 30
 - Web: 10
 - Top flange width: 197
 - Bottom flange width: 150
 - Splice on side 1: 315
 - Splice on side 2: 315
 - Splice support: 472.5
 - Diaphragms: 150
 - Bottom web support: 150

Bottom web support

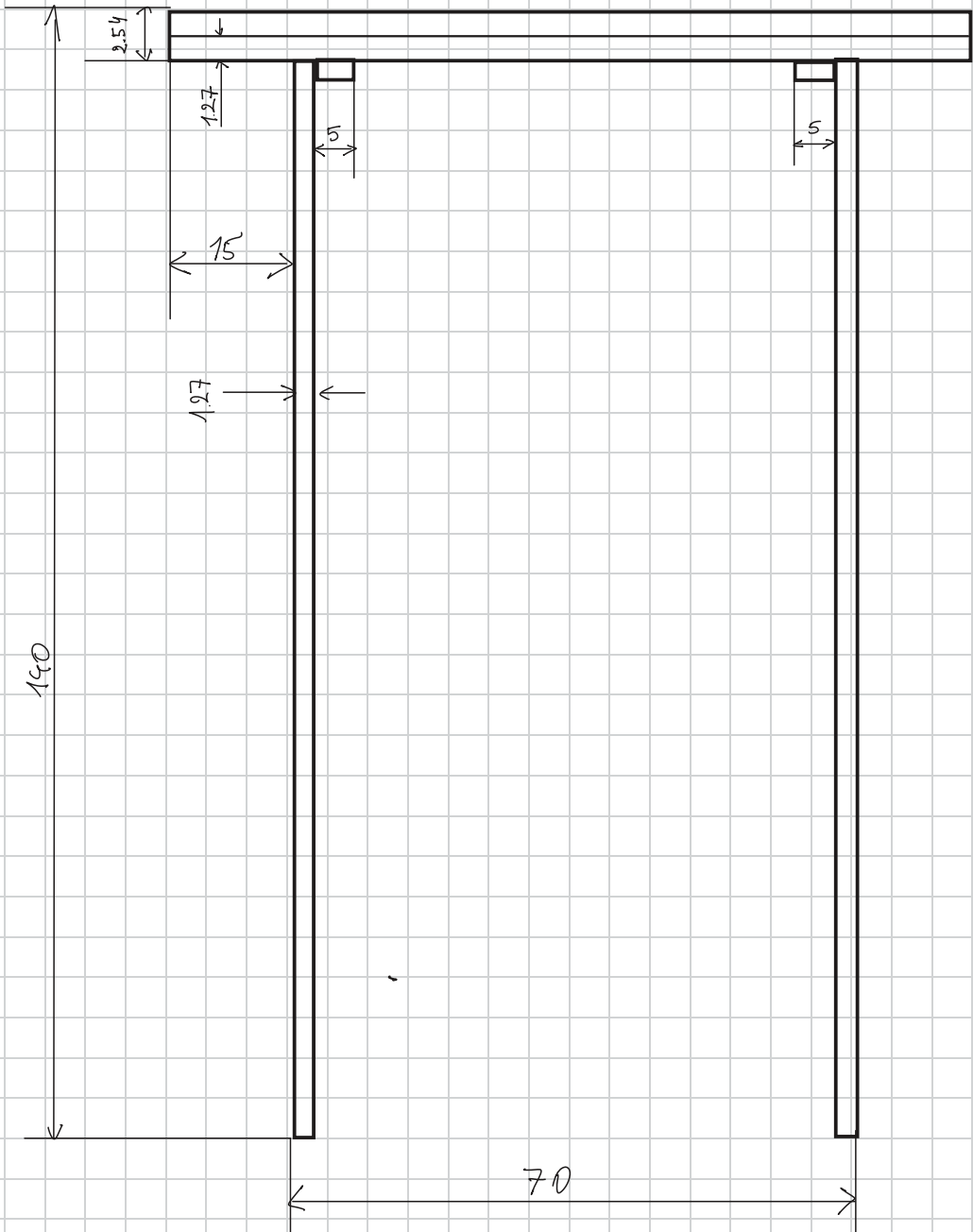
Bridge Top view



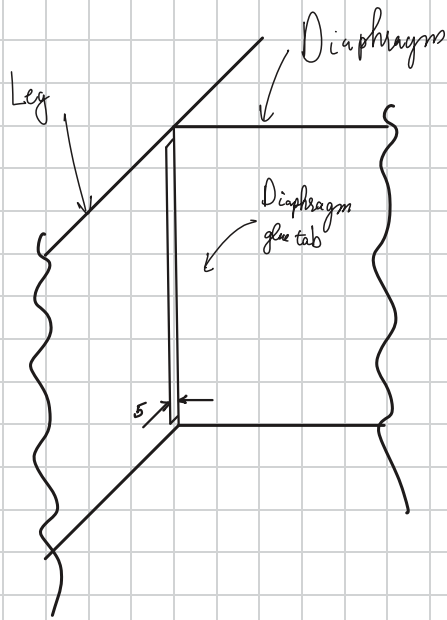
Bridge Bottom view



Bridge cross section view



Diaphragm joints



Construction Section:

Precautions:



Figure 1 (November 22, 2025, 5:06 PM)

Team members wearing disposable gloves and surgical masks during construction to protect against contact cement fumes and prevent skin exposure.

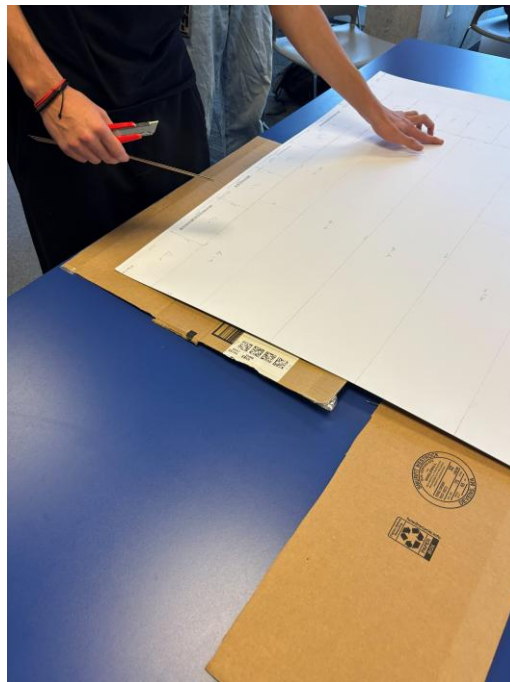


Figure 2 (November 21, 2025, 1:13 PM)

A protective cardboard sheet placed underneath the matboard while cutting, preventing damage to the table surface and ensuring safe, controlled cutting operations.



Figure 3 (November 22, 2025, 5:39 PM)

Workspace prepared for applying contact cement, with gloves being worn and a secondary cardboard layer used to prevent glue spills from touching the table surface.

Building space after finishing tasks

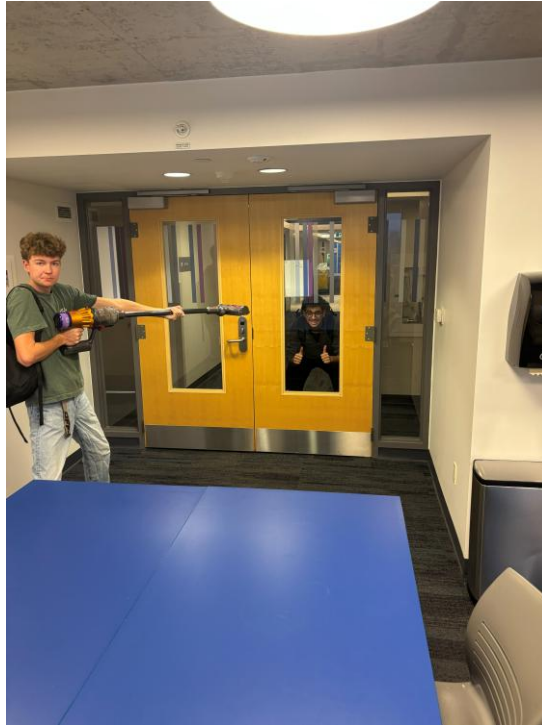


Figure 4 (November 21, 2025, 3:18 PM)

Building space after completing the cutting stage. The area is cleaned of loose matboard pieces and cutting tools are organized.



Figure 5 (November 22, 2025, 8:49 PM)

Building space after the gluing stage, showing all materials stored safely and the surface cleaned of excess contact cement.

Building stages



Figure 6 (November 21, 2025, 12:49 PM)

Carefully tracing guidelines on the matboard to ensure precise cuts and folds, forming the foundation for accurate assembly of the bridge geometry.

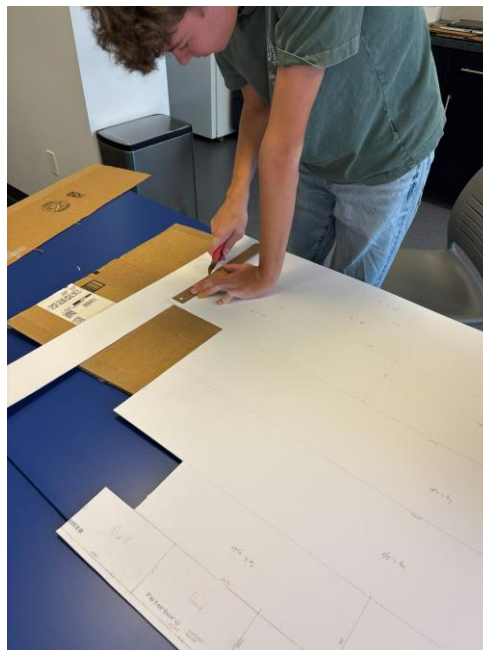


Figure 7 (November 21, 2025, 2:52 PM)

Cutting the matboard pieces according to the predefined dimensions, using a straightedge and utility knife over a protective cardboard layer to avoid damaging the table.

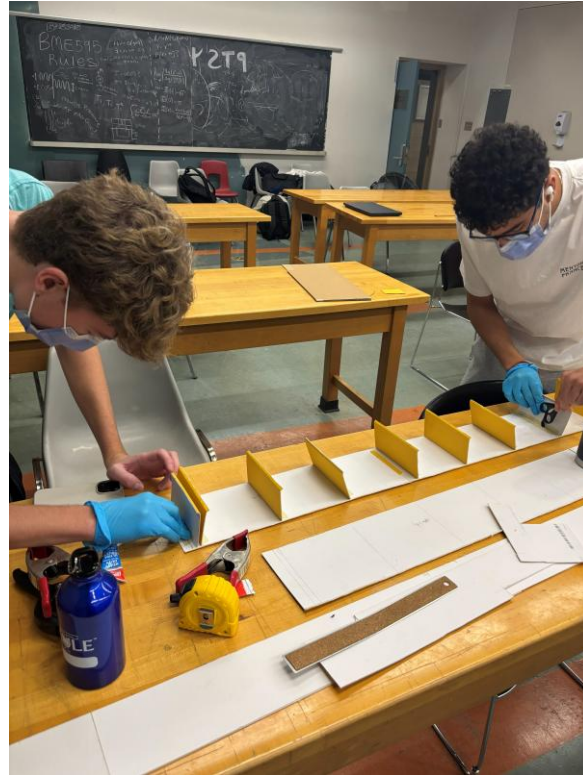


Figure 9 (November 22, 2025, 7:06 PM)

Gluing major structural components using contact cement, with protective gloves and masks worn.



Figure 9 (November 23, 2025, 1:17 PM)

Applying a uniform pink spray paint coat to the bridge components to match the team's colour theme. Painting was performed in a well-ventilated area while wearing masks and gloves for safety.



Figure 10 (November 23, 2025, 6:02 PM)

The fully constructed and painted final bridge, featuring a uniform pink finish and decorative drawings. This image shows the completed structure meeting all dimensional constraints and ready for transport to the CIV102 testing session.

Time log:

Tasks	Date	Oliver	Tristan	Yassine	Total
Deliverable 1					
Part 1	Nov 5	0.5	–	–	0.5
Part 2	Nov 7	–	0.5	–	0.5
Part 3	Nov 8	–	–	0.5	0.5
Team Time Log Meeting	Nov 10	0.5	0.5	0.5	3
Coding					
Initial coding	Nov 12–16	8	1	1	10
Design 0 Hand Calculations	Nov 12–16	–	2	2	4
Code–Hand Calculation Check	Nov 18	0.5	0.5	0.5	1.5
Iteration to Final Design	Nov 19	3	3	3	9
Design Report – Key Points	Nov 20	–	3	–	3
Engineering Drawings	Nov 20	–	–	3	3
Check-in Meeting (Code + Iteration + Drawings)	Nov 21	2	2	2	6
Cutting Pieces	Nov 22	3	3	3	9
Final Bridge Construction	Nov 23	5	5	5	15
Total (hrs)		22.5	20.5	20.5	

Signature 1 : Yassine Lahbichi

Signature 2 : Oliver de Boer

Signature 3 : Tristan Steyn